

# FABIAN D. HERRERA

fabianherrerap73@gmail.com | +1 (438) 630-5373 | [Github](#) | [Linkedin](#) | [Portfolio](#)

## EDUCATION

### McGill University

B.Sc. in Software Engineering - Minor in Entrepreneurship

845 Sherbrooke St W, Montreal, QC

Expected Graduation: 2028

- **Relevant coursework:** Data Structures and Algorithms, Software Systems, Algorithms and Complexity, Computer Systems, Probability and Statistics.

## PROFESSIONAL EXPERIENCE

### Transport Laberge

Data Engineering & Analytics Intern

Montreal, QC

January 2026 – June 2026

- Built scalable **ETL pipelines** processing **20M+ records** using **Python** and **SQL**, removing manual ingestion steps and enabling reliable, production-ready ML data flows.
- Working on an **AI voice agent** that connects **OpenAI APIs** with internal transport systems to answer operational questions, extract logistics information, and trigger workflows, with the goal of reducing dispatcher manual workload by **at least 25%** by the end of the internship.
- Operated in an Agile workflow with daily stand-ups, delivering targeted features each sprint and tracking progress in Azure DevOps.
- Built **Power BI** reports visualizing key KPIs (FTFR, MTBF, TCO) to drive data-based decisions.
- Deployed an API on **Vercel** (Vapi, Supabase) to automate post-call email workflows.

## PROJECTS & RECOGNITIONS

### Logbait™ – Cloud-Integrated Roguelike Game | [Play](#)

Team Project (8) | McGill

Montreal, QC

January 2026 – May 2026

- Shipped a **browser-playable** action roguelike to itch.io with a team of 8, by owning the core gameplay loop in **Unity** and **C#**: player movement, combat resolution, enemy AI targeting, and the health and damage system.
- Enabled teammates to add new player powers without touching combat code, cutting integration conflicts across an 8 person team, by designing a **modular ability system** in **C#** with reusable cooldown, VFX, and state handling.
- Kept fast combat readable under load by building a responsive combat UI driven off combat state events

### Swap - P2P Currency Exchange Web App | [GitHub](#)

Personal Project

Montreal, QC

July 2025 – Now

- Shipped a live peer to peer currency exchange platform by building a **React 19** and **Vite** frontend on **Vercel** against an **Express 5** API on **Render** and **MongoDB Atlas**, secured with **JWT** auth and protected routes.
- Cut offer creation from a full form to one typed sentence by parsing currencies, amounts, and direction with **deterministic rules first**, calling the **Gemini API** only on ambiguity within an 8 second budget.
- Made AI output safe to ship by revalidating every model response against the manual form's rules, so the feature **degrades to a partial fill** instead of failing on timeout or rate limit.
- Protected users from contact scraping by releasing emails **server side only after an accept**, atomically auto declining competing requests.

### House Price Prediction Model | [Certificate](#) | [Car Model](#)

Dawson AI Research and Development Project

3040 Sherbrooke St W, Montreal, QC

August 2023 - July 2024

- Developed a linear regression model to predict used car prices with **~80% accuracy (R-squared)**.
- Cleaned **50,000+ entries**, cutting error rates by **20%**, handling missing values and outliers.
- Evaluated the model using **MSE** and **MAE**, achieving a low MSE indicating accurate predictions.
- Built with **Python** using **Pandas**.
- Collaborated with my team to develop a house price prediction model using linear regression techniques.

## TECHNICAL SKILLS

**Programming Languages:** Python, JavaScript, Java, C#, HTML, CSS, R, Bash/sh

**Frameworks & Libraries:** React, Node.js, Express, Mongoose, Bootstrap, Pandas, NumPy, scikit-learn

**Database:** MongoDB

**Tools & Technologies:** Git/Github, Jupyter Notebook, Unity, Visual Studio Code, Canva, Figma, Azure DevOps, Power BI, Power Automate, SharePoint, Vercel, Supabase, MS Office 365

**Concepts:** Object-Oriented Programming (OOP), REST APIs, Machine Learning (Linear Regression), Data Preprocessing, Model Evaluation (MSE, MAE), Bash/sh scripting and Linux, Agile/Scrum, CI/CD, DevOps, ETL, KPI Reporting.

**Languages:** French (Fluent), English (Fluent), Spanish (Fluent)

**Others:** Possess a valid driver's license and own a car.